Math 10C — Unit 3: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Vocabulary, Terms and Degree

1. Identify as a monomial, a binomial, or a trinomial.
(a) $y + 5$ (b) $7y^4$ (c) $3y^2 + 5y + 1$

2. State the degree of each monomial.
(a) $8b$ (b) $4x^2y$ (c) 15 (d) $-5m^3n^2$ (e) $4pq^3r^2$

3. State whether or not the following are polynomial expressions. If they are not, explain why not.
(a) $\frac{1}{3}x^2 - 5x$ (b) $6n^{-3}$ (c) $\sqrt{10}$

(d) $\frac{9}{x^4}$ (e) $\frac{5x^3}{4}$ (f) $x^5 + 4x^{2.5}$

4. Complete the following table for the single-variable polynomials.

Polynomial Expression	Leading Coeff.	Constant Term	Degree
$t^4 - t + 9$			
$0.3s^3 - 0.2s^2 + 0.5s - 0.4$			
$\sqrt{7} - x^5$			
$2\pi x^2 - 9 - 5x$			
$-\frac{1}{8}c^2$			

5. Give an example of:
(a) a trinomial of degree 3 in one variable.

(b) a binomial in five variables with a constant term of 9.

(c) a monomial of degree 4 in two variables with a negative numerical coefficient.

(d) a monomial with a degree of 0.

Part B — Ordering, True/False and Tiles

6. Arrange the following in descending powers of the variable.

(a) $\frac{1}{5}a^2 - \frac{3}{4}a^3 - 2 - a$

(b) $z - 5 - 3z^7 + z^4$

7. Arrange the following in ascending powers of the variable.

(a) $4x^2 - 3x^6 - 5x^5 - 3x^4 + 7x - 9$

(b) $6x^3 - 6x + 6$

8. State which of the following are true and which are false.

(a) -78 is a polynomial.

(b) The degree of the polynomial $4x^4y^4$ is 8.

(c) The numerical coefficient of $\frac{8x}{9}$ is 8.

(d) A polynomial may have 5000 terms.

(e) $\frac{3}{b^4} - 1$ is a binomial.

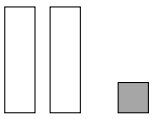
(f) The degree of the polynomial 7 is 0.

(g) The polynomial $2x^3 + 5x^2 + 4x + 9$ is written in ascending powers of x.

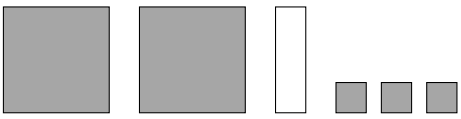
(h) The polynomials $4x^2 - 7x + 2$ and $2 - 7x + 4x^2$ are equivalent.

9. State the polynomial expression which describes each diagram.

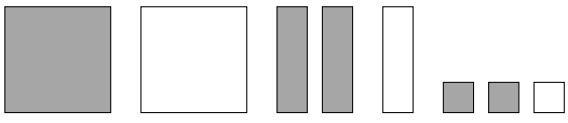
(a)



(b)



(c)



Part C — Adding and Subtracting Polynomials

10. Use algebra tiles (or reasoning about zero pairs) to determine the result of the addition of:
(a) $(x^2 - 2x - 4) + (x^2 - 3x - 2)$ (b) $(5x + 3) + (3x^2 - 5x - 5)$

11. Use algebra tiles (or reasoning about zero pairs) to determine the result of the subtraction of:
(a) $(x^2 - 5) - (3x^2 + 2x + 3)$ (b) $(5 - 2x - x^2) - (2 - 4x + x^2)$

12. Simplify.
(a) $\frac{1}{3}y - 5 + \frac{5}{3}y + 20$ (b) $5b^3 + 9b - 3b^2 - 8b - 5b^3 - 2b^2$ (c) $5 - 3x + 6y + 5y - 3x + 9z - 12$

13. Simplify the following polynomial expressions by collecting like terms.
(a) $(6p - 8q - 3r) + (r - 9q - 2p)$ (b) $(3x^2 + 4x - 2) + (2x - 5 - 5x^2) + (3 - 4x + 2x^2)$

(c) $(5p - 7q) - (6p - 3q)$ (d) $\begin{array}{r} 8x^2 + 3x - 2 \\ -\ (-4x^2 - 4x - 2) \\ \hline \end{array}$ (e) $\begin{array}{r} -5x^2 + 3x - 7 \\ -\ (4x + 5 - 3x^2) \\ \hline \end{array}$

14. Subtract as directed.
(a) Subtract $4x^2 - 3x + 5$ from $9x^2 - 7x - 1$.
(b) Subtract the sum of $(3x^3 - 5x^2 - 4x + 2)$ and $(6 - 2x + 4x^2 - 5x^3)$ from $(3x^3 - 8x + 7)$.

15. A triangle has a perimeter of $(7m + 2n)$ cm. One side measures $(3m - 4n)$ cm and another side measures $(4n + 2m)$ cm.
(a) Write and simplify an expression for the length of the third side of the triangle.
(b) Determine the measure of each side when $m = 5$ and $n = -2$.

Part D — Multiple Choice and Numerical Response

16. *(Multiple Choice)* Which of the following is a polynomial expression of degree 4?

A. $5x^5 - 3x^2$

B. $6x^4 - 2x^3 + 3x^{-1} + x - 4$

C. $\dfrac{6x^4 - 5x}{x}$

D. $7 + 4x - \dfrac{2}{5}x^2 - x^3 + \dfrac{3}{7}x^4$

17. *(Multiple Choice)* Which of the following polynomial expressions, when simplified, is equal to $6x$?

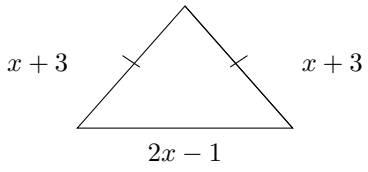
A. $(4x^2 - 4x) - (3x + 4x^2)$

B. $6x - (3x^2 - 3x) + (3x^2 + 3x)$

C. $9 + (5 - 3x) - (14 - 9x)$

D. $(3x^2 - 4x + 5) - (3x^2 - 4x) + (9x - 5)$

18. *(Multiple Choice)* The perimeter of the isosceles triangle shown can be represented by A. a monomial B. a binomial C. a trinomial D. none of the above



19. *(Numerical Response)* If the polynomial $9 - 4x + 5x^2 - 7x^5$ has degree a, leading coefficient b, and constant term c, then the value of $2a - 3b + c$ is _____.

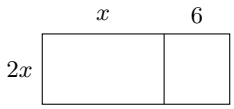
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Lesson 2 Assignment *Show your work.*

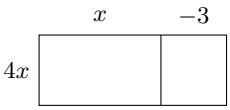
Part A — Area Models and the Distributive Property

1. State the polynomial product represented in each area model.

(a)



(b)



2. For each of the following: i) sketch an area model, ii) express the product as a sum or difference of terms, and iii) verify the result when $x = 6$.

(a) $3x(2x + 3)$

(b) $-2x(5 - x)$

3. Expand.

(a) $5(6x - 2)$

(b) $-3(3x + 7)$

(c) $5x(3y + 4z)$

(d) $-x(x - 6y)$

(e) $4(x - 3y + 2z)$

(f) $-3a(2b - c + 4d)$

(g) $(x + 4)5x$

(h) $3x(x - 4y + 2z)$

4. Expand and simplify.

(a) $4(x + 7) - 9$

(b) $12 - 3(4x + 9)$

(c) $5(x - 3) + 2x$

(d) $3(x + 4) + 5(3x - 2)$

(e) $3(x + 4) - 5(3x - 2)$

(f) $-3(x + 2) + 6(2x - 1)$

(g) $4(-x + 9) + 4(x - 6)$

5. Identify the error in each simplification below and give the correct simplification.

(a) $4x(3x + y) = 12x + 4xy$

(b) $x^3(x^2 - 3x + 5) = x^6 - 3x^4 + 5x^3$

(c) $3(x-4)-3(x-2) = 3x-12-3x-6 = -18$

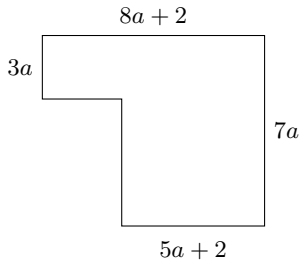
6. Expand and simplify.

(a) $3a(a+5) - 2a(4a-1)$ (b) $5(x^2+2) - (3x^2-1)$ (c) $3(x+4) - x - 2$

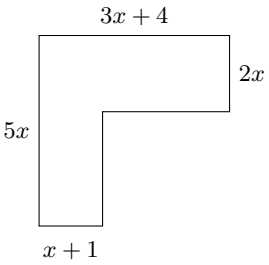
(d) $w(w^2+5) - (2w+9)$ (e) $4(9x-2y) + 3(3y+x)$ (f) $-3x(x^3+2x^2) - 5x(3x^3-x^2)$

7. Determine a simplified expression for the area of each shape.

(a)



(b)



8. *(Multiple Choice)* The area model below represents the expansion of

A. $3x(x+2)$ B. $-3x(x+2)$ C. $3x(x-2)$ D. $-3x(x-2)$

	x	-2
$3x$	$+3x^2$	$-6x$

9. *(Multiple Choice)* Which of the following expansions is *incorrect*?

A. $-3x^2(2x+5) = -6x^3 - 15x^2$ B. $-5x(3-x) = -15x + 5x^2$

C. $-4x(x^2-6) = -4x^3 - 24x$ D. $6x^2(x^2+4) = 6x^4 + 24x^2$

10. *(Numerical Response)* The expression $3x(5-2x) + 4x(x-3) - 2(3x+1)$ can be written in the form $ax^2 + bx + c$. The value of $c - a - b$ is _____.

Record your answer left to right. | | | | |

Part B — Multiplying Two Binomials

11. Use an area model to determine each product.

(a) $(x+4)(x+6)$

(b) $(3x+2)(2x+5)$

(c) $(y-6)(5y+3)$

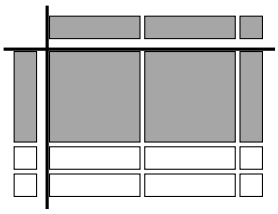
(d) $(a^2+4)(a^2-9)$

12. Use algebra tiles to model each product. Sketch the factor tiles across the top and down the side of a frame, fill in the rectangle using the shading rule, and state the product.

(a) $(x+1)(2x+3)$

(b) $(x+3)(x-2)$

13. The algebra-tile model below shows a product of two binomials.



(a) What two factors are shown along the top and the side?

(b) State the product, simplified.

14. Use an area model and no calculator to determine each product.

(a) 34×22

(b) 41×63

(c) 58×27

(d) 76×76

15. **(a)** Expand $(x+3)(x+4)$. **(b)** Substitute $x = 10$ into both $(x+3)(x+4)$ and your expansion to calculate 13×14 without a calculator. **(c)** Explain why this works.

16. Use the distributive property to determine each product.
(a) $(x+6)(x+9)$ (b) $(a+8)(4a-3)$

(c) $(p-5)(q-9)$ (d) $(x+7y)(x-3y)$

17. Use FOIL to determine each product.
(a) $(x+4)(x+8)$ (b) $(y+9)(y-5)$ (c) $(m-3)(m-11)$

(d) $(2x-7)(3x+4)$ (e) $(5a-2b)(a+6b)$ (f) $(7-3y)^2$

18. A rectangle has length $(4a+3)$ cm and width $(a+6)$ cm. Determine the area of the rectangle, in cm^2, by completing each of the following.
(i) use an area model (ii) use the distributive property (iii) use FOIL

19. Expand and simplify where possible.
(a) $(5x-3)(2x+7)$ (b) $(3h-4)(3h+1)$

(c) $(2z+5)(2z-5)$ (d) $(x^2+3y^2)(2x^2-4y^2)$

20. *(Numerical Response)* The area of a rectangle with length $(3x+1)$ cm and width $(x+2)$ cm can be written in the form px^2+qx+r, where p, q, r are natural numbers. Write the value of p in the first box, q in the second box, and r in the third box.

Record your answer left to right. | | | | |

21. *(Numerical Response)* The expansion of $(4x-c)(x-5)$, where c is a whole number, results in a polynomial in x with a leading coefficient of 4 and a constant term of 20. The value of c is _____.

Record your answer left to right. | | | | |

Lesson 3 Assignment *Show your work.*

Use the two shortcuts wherever they apply, and expand every product completely before collecting like terms.

1. Expand and simplify where possible.

(a) $(x-6)^2$ (b) $(x-11)(x+11)$ (c) $(4x-y)^2$

(d) $(3x+5y)^2$ (e) $(4x-3)(4x+3)$ (f) $(-3y+2)^2$

(g) $(5p+4)^2$ (h) $(6m+5n)(6m-5n)$ (i) $(7a-2b)^2$

(j) $(8-3x^2)(8+3x^2)$ (k) $(5a-9b)(5a-9b)$ (l) $(3a^3-4)(3a^3+4)$

2. Expand and simplify where possible.

(a) $3(2x-5)(4x-1)$ (b) $4(3x-1)(5x+2)$ (c) $-2(a+6)(3a+7)$

(d) $3(5x+2)^2$ (e) $5(6x-2y)(3x+y)$ (f) $-6(a+4b)(3a-2b)$

3. Expand and simplify where possible.

(a) $(x-4)(x-7)+(x+3)(x+6)$ (b) $(x-6)(x+5)-(x+2)(x-9)$

(c) $(x-4)^2+(x+4)^2$ (d) $(x-y)(x-5y)-(x+y)(x-y)$

4. Expand and simplify where possible.

(a) $(2x-1)(x-4)-3x(x-2)$ (b) $(3x+2)(2x+1)-(2x-5)(3x-4)$

(c) $(7x-1)(x-3)-(2x+1)(2x-1)$ (d) $6(4-2x)(3+4x)-2(1+x)^2$

(e) $4(3x-2)(3x+4)+2(x+5)(x+3)$ (f) $3(3p+2q)^2-(4p-q)(5p+7q)$

5. Expand and simplify where possible.

(a) $(x+5)^2 + (x+3)^2$

(b) $(4a-b)^2 - (3a+4b)^2$

(c) $4(y-2)^2 - 3(2y-1)^2$

(d) $7 - 3(x-2)(x+5) + (2x-3)(x-4)$

(e) $4(1+2y)(3-y) - (2y-3)(2y-4)$

6. *(Multiple Choice)* A student provides the following expansions for four binomial products.

$(x+4)^2 = x^2 + 16$ $(2x-y)(2x-y) = 4x^2 - 4xy + y^2$ $(3x+y)(3x-y) = 9x^2 - y^2$ $(4x+5)^2 = 16x^2 + 40x + 25$

How many of the student's expansions are *incorrect*? A. One B. Two C. Three D. Four

7. A square patio has side length x m. Each side length of the patio is increased by 3 m.

(a) Write and simplify an expression for the area of the new patio.

(b) Write and simplify an expression for the *increase* in area.

(c) Check your answer to (b) for a patio that was 5 m by 5 m.

8. Use substitution to check two expansions of $(3x-4)^2$.

(a) Verify that $(3x-4)^2 = 9x^2 - 24x + 16$ by substituting $x = 2$ into both sides.

(b) Jen says $(3x-4)^2 = 9x^2 + 16$. Use $x = 2$ to show that Jen is wrong, and explain her error.

9. *(Extension) (Numerical Response)* Given that, for every value of x, $x^2 - 14x + 55 = (x-a)^2 + b$, then the value of b must be _____.

Record your answer left to right. | | | | |

Lesson 4 Assignment *Show your work.*

1. Use an area model (grid) to determine the expansion.
(a) $(p-4)\left(p^2+3p+5\right)$ (b) $(2n+5)\left(n^2-4n+3\right)$

2. Use the distributive property to determine the expansion.
(a) $(x-6)\left(x^2-2x+7\right)$ (b) $(3a+4)\left(a^2-5a-6\right)$

(c) Check your answer to (a) by substituting $x=2$ into both $(x-6)\left(x^2-2x+7\right)$ and your expansion.

3. Marco tries to expand $(x+2)\left(x^2+x+1\right)$ with FOIL and gets x^3+x+2x^2+2.
(a) Explain why FOIL alone cannot expand this product. (b) Expand $(x+2)\left(x^2+x+1\right)$ correctly.
Which partial products did Marco miss?

4. Expand and simplify.
(a) $\left(x^2-5\right)\left(3x^3+2x-9\right)$ (b) $\left(-n^2-n+2\right)(n+3)$

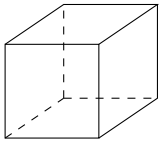
(c) $(a-4b)\left(3a^2-2ab-5b^2\right)$ (d) $3(4x+3)\left(2x^2+x-5\right)$

5. Expand and simplify.
(a) $(x+3)(x+5)(2x+1)$ (b) $(h-5)(3h-2)(2h-1)$

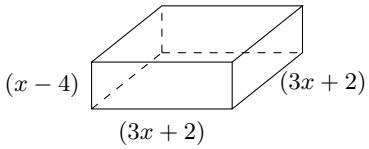
(c) $(a+2b)(3a-4b)(3a+4b)$

(d) $(2x+5y)(3x-2y)(x-5y)$

6. Calculate the volume of the cube shown below.



7. Calculate the volume of the rectangular prism illustrated.



8. Simplify.

(a) $-4\left(b^2+3\right)\left(2b^2-b-1\right)$

(b) $\left(-3x^2-2x+1\right)\left(x^2-x-4\right)$

(c) $3(2x-1)^2-(x-3)^3$

9. *(Numerical Response)* Subtracting the product of $(2x-1)$ and $\left(3x^2-5x+2\right)$ from the sum of $\left(3x^3-4x^2-7\right)$ and $\left(x^2+5x-2\right)$ results in a polynomial of the form ax^3+bx^2+cx+d. The value of $b-2c$ is _____.

Record your answer left to right. | | | | |

10. *(Multiple Choice)* A student attempts to expand $(b+3)^3$. Her work is shown below.

$$\begin{aligned}
&(b+3)^3 \\
&= (b+3)(b+3)(b+3) && \text{Line 1} \\
&= (b+3)\left(b^2+9\right) && \text{Line 2} \\
&= b^3+3b^2+9b+27 && \text{Line 3}
\end{aligned}$$

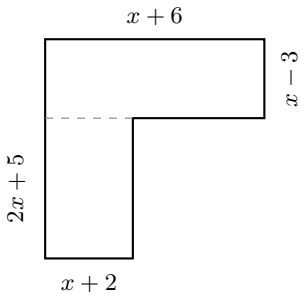
Which of the following statements is true?

A. The student made an error in Line 1.
B. The student made an error in Line 2.
C. The student made an error in Line 3.
D. The student's expansion is correct.

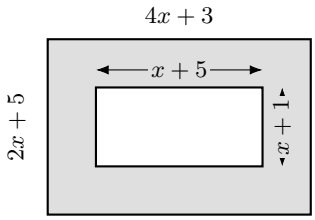
Lesson 5 Assignment *Show your work.*

1. In each case, the figure consists of a series of horizontal and vertical lines. The area of each figure can be written as a trinomial in the form $ax^2 + bx + c$. Determine the values of a, b, and c, and calculate the area for the given value of x.

(a) A figure formed from a rectangle of width $(x + 2)$ and height $(2x + 5)$, joined along its top edge to a second rectangle of width $(x + 6)$ and height $(x - 3)$. Evaluate the area when $x = 5$.

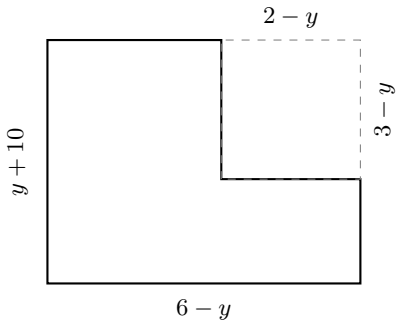


(b) A rectangular frame with outer dimensions $(4x+3)$ by $(2x + 5)$ and a rectangular opening of dimensions $(x + 5)$ by $(x + 1)$ cut from its interior. Evaluate the area when $x = 1.5$.



2. A figure consists of a rectangular bounding box of width $(6 - y)$ and height $(y + 10)$, with a rectangular notch of width $(2 - y)$ and height $(3 - y)$ removed from one corner.

(a) Determine the area of the figure in the form $ay^2 + by + c$.



(b) Determine the area of the figure when $y = -3$.

3. A square metal plate of side 30 cm is heated so that each side increases in length by x cm.

(a) Write and simplify an expression for the area of the heated plate.

(b) Write and simplify an expression for the increase in area of the plate.

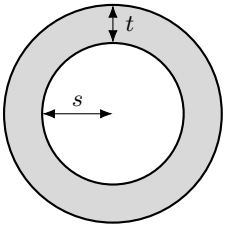
(c) If $x = 0.3$, calculate the increase in area.

4. A square garden with side length $(4x + 1)$ m contains two square flower beds, each with side length $(x + 2)$ m. The remainder of the garden is grass.

(a) Draw a diagram to illustrate this information. *(Hint: draw a large square and label each side $4x + 1$; draw two smaller squares inside it and label each side $x + 2$.)*

(b) Write and simplify an expression for the area of grass in the garden.

5. A metal washer has internal radius s mm and width t mm.



(a) Write an expression for the outer radius of the washer.

(b) Show that the area of the washer, A mm^2, is given by $A = 2\pi st + \pi t^2$.

6. Solve the following equations where the variable is a real number.

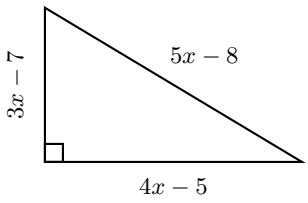
(a) $(2y - 1)(y - 3) = 2y(y - 4)$

(b) $(t + 3)^2 = t^2 + 15$

(c) $m^2 - (m - 7)^2 = 21$

(d) $2n^2 - (n - 4)^2 = (n + 3)(n - 2)$

7. The hypotenuse of a right triangle is $(5x - 8)$ cm long, and the lengths of the other two sides are $(4x - 5)$ cm and $(3x - 7)$ cm. Form an equation and solve it to determine the value of x and the lengths of the three sides of the triangle.



8. Consider a set of rectangles with sides $(5a - 2)$ cm and $(3a + 4)$ cm.

(a) Write and simplify an expression in a for the area of one of these rectangles.

(b) If one of these rectangles has a perimeter of 60 cm, determine the length and width of this rectangle.

(c) If another of these rectangles is a square, determine the length of each side.

9. A rectangle has length $(x^2 + 5x - 2)$ cm and width $(2x - 3)$ cm.

(a) Write and simplify an expression for the area of the rectangle in cm^2.

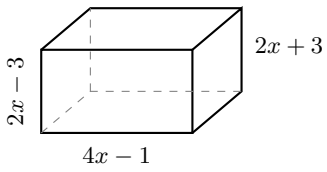
(b) If $x = 2.5$, calculate the area of the rectangle.

10. Dice for a children's board game are cubes with an edge length of $(2x - 1)$ mm.

(a) Write and simplify an expression for the volume of a die in mm^3.

(b) The manufacturer packages dice in cubic containers containing 27 dice. Determine the volume of the container in cm^3 if $x = 8$.

11. A rectangular prism has length $(4x - 1)$ cm, width $(2x + 3)$ cm, and height $(2x - 3)$ cm.



(a) Write and simplify an expression for the volume of the rectangular prism in cm^3.

(b) Write and simplify an expression for the surface area of the rectangular prism in cm^2.

(c) If $x = 3$, calculate the volume and surface area of the rectangular prism.

12. *(Multiple Choice)* A box is in the shape of a rectangular prism. The length of the box is y cm. The width is 3 cm less than the length, and the height is 5 cm more than the length. If the volume of the box can be written in the form $V = ay^3 + by^2 + cy + d$ where a, b, c, d are integers, then how many of the parameters a, b, c, d are equal to zero?

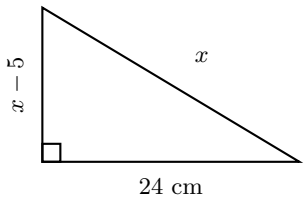
A. 0 B. 1 C. 2 D. 3

13. *(Numerical Response)* A square has side length x cm. A rectangle has side lengths $(x - 2)$ cm and $(x + 7)$ cm. If the square and the rectangle are equal in area, the value of x, to the nearest tenth, is _____.

Record your answer left to right. | | | | |

14. *(Numerical Response)* A right triangle has one leg of length $(x - 5)$ cm, the other leg of length 24 cm, and a hypotenuse of length x cm. The perimeter of the triangle, to the nearest centimetre, is _____ cm.

Record your answer left to right. | | | | |



Lesson 6 Assignment *Show your work.*

The questions are not in the same order as the examples — decide which skill each one needs.

Part A — Multiple Choice

1. The degree of the polynomial $2a^2b^3 - 7ab + 4b^4$ is A. 3 B. 4 C. 5 D. 11

2. Which of the following are polynomials of degree 4?

I. $6x^4 - x$ II. $3x^4 - 2x^5$ III. $x^2y^2 + 5$ IV. $2x^3 - 5x^{-1} + x^4$

A. I only B. I and III only C. I, III and IV only D. I, II, III and IV

3. Which product, when expanded and simplified, has exactly *two* terms?

A. $(3x + 5)(5 - 3x)$ B. $(x + 5)^2$ C. $(x - 5)(x + 3)$ D. $(x - 5)(5 - x)$

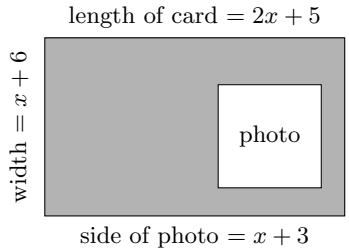
4. Exactly one of the following expansions is *incorrect*. Which one?

A. $(x - 4)(x + 6) = x^2 + 2x - 24$
B. $(3x - 2)^2 = 9x^2 - 6x + 4$
C. $(2x + 3)(x - 5) = 2x^2 - 7x - 15$
D. $(x + 7)(x - 7) = x^2 - 49$

5. The area, in m^2, of a rectangle with length $(2k - 3)$ m and width $(k + 4)$ m is

A. $2k^2 + 5k - 12$ B. $2k^2 - 12$ C. $2k^2 + 11k - 12$ D. $2k^2 + 5k + 12$

6. *Use the following diagram.* A square photo with side length $(x + 3)$ cm is glued onto a rectangular card that is $(2x + 5)$ cm long and $(x + 6)$ cm wide.



The area of the card, in cm^2, that is *not* covered by the photo is

A. $x^2 + 11x + 39$ B. $x^2 + 17x + 21$ C. $3x^2 + 23x + 39$ D. $x^2 + 11x + 21$

7. Expand $(2a-1)(a+3)(a-2)$.

A. $2a^3+6$ B. $2a^3+3a^2-13a+6$ C. $2a^3+a^2-13a-6$ D. $2a^3+a^2-13a+6$

8. A student expands $(x-3)\left(x^2+2x-1\right)$. Her work is shown below.

$$\begin{aligned}
&(x-3)\left(x^2+2x-1\right)\\
&= x\left(x^2+2x-1\right)-3\left(x^2+2x-1\right) && \text{Line 1}\\
&= x^3+2x^2-x-3x^2-6x+3 && \text{Line 2}\\
&= x^3-x^2-5x+3 && \text{Line 3}
\end{aligned}$$

Which of the following statements is true?

A. The student made an error in Line 1.
B. The student made an error in Line 2.
C. The student made an error in Line 3.
D. The student's expansion is correct.

Part B — Numerical Response

9. When $(x+2)^2-(x+4)(x-4)$ is expanded and simplified, it can be written in the form $ax+b$. The value of $a+b$ is _____.

Record your answer left to right. | | | | |

10. A sheet of cardboard measures 20 cm by 14 cm. A square of side x cm is cut from each corner, and the sides are folded up to make an open box. When $x=3$, the volume of the box, in cm^3, is _____.

Record your answer left to right. | | | | |

Part C — Written Response

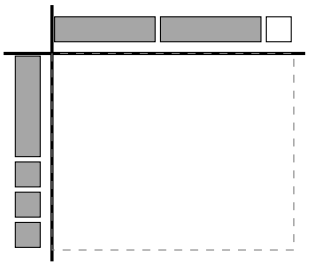
11. *(5 marks)*

(a) Expand and simplify $(2x-5)(3x+4)$.

(b) Expand and simplify $(3x-2)\left(x^2+4x-5\right)$.

(c) Verify your answer to (b) by substituting $x=2$ into both the original product and your expansion.

12. *(3 marks)* The factor tiles for a product are shown across the top and down the side of the frame.



(a) State the two factors.

(b) Complete the tile rectangle using the shading rule, and state the product, simplified. Explain how zero pairs are used.

Part D — Extension (not part of the core Math 10C review)

13. *(Extension) (Numerical Response)* The trinomial $x^2 - 18x + 85$ can be expressed in the form $(x - p)^2 + q$. Write the value of p in the first box and the value of q in the second box.

Record your answer left to right. | | | | |

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. (a) binomial (b) monomial (c) trinomial. **2.** (a) 1 (b) 3 (c) 0 (d) 5 (e) 6. **3.** (a) yes (b) no — negative exponent (c) yes, a constant polynomial (d) no — $\frac{9}{x^4} = 9x^{-4}$, negative exponent (e) yes (f) no — the exponent 2.5 is not a positive integer. **4.** $t^4 - t + 9$: LC 1, CT 9, deg. 4; $0.3s^3 - 0.2s^2 + 0.5s - 0.4$: LC 0.3, CT -0.4, deg. 3; $\sqrt{7} - x^5$: LC -1, CT $\sqrt{7}$, deg. 5; $2\pi x^2 - 9 - 5x$: LC 2π, CT -9, deg. 2; $-\frac{1}{8}c^2$: LC $-\frac{1}{8}$, CT 0, deg. 2. **5.** Answers may vary: (a) $x^3 + x^2 - 4$ (b) $pqrst + 9$ (c) $-5m^2n^2$ (d) 12. **6.** (a) $-\frac{3}{4}a^3 + \frac{1}{5}a^2 - a - 2$ (b) $-3z^7 + z^4 + z - 5$. **7.** (a) $-9 + 7x + 4x^2 - 3x^4 - 5x^5 - 3x^6$ (b) $6 - 6x + 6x^3$. **8.** (a) true (b) true (c) false (d) true (e) false (f) true (g) false (h) true. **9.** (a) $-2x + 1$ (b) $2x^2 - x + 3$ (c) $x + 1$. **10.** (a) $2x^2 - 5x - 6$ (b) $3x^2 - 2$. **11.** (a) $-2x^2 - 2x - 8$ (b) $-2x^2 + 2x + 3$. **12.** (a) $2y + 15$ (b) $-5b^2 + b$ (c) $-6x + 11y + 9z - 7$. **13.** (a) $4p - 17q - 2r$ (b) $2x - 4$ (c) $-p - 4q$ (d) $12x^2 + 7x$ (e) $-2x^2 - x - 12$. **14.** (a) $5x^2 - 4x - 6$ (b) $5x^3 + x^2 - 2x - 1$. **15.** (a) $(2m + 2n)$ cm (b) 23 cm, 2 cm and 6 cm. **16.** D. **17.** C. **18.** B. **19.** 40.

Lesson 2 Assignment

1. (a) $2x(x + 6) = 2x^2 + 12x$ (b) $4x(x - 3) = 4x^2 - 12x$. **2.** (a) $6x^2 + 9x$ (b) $2x^2 - 10x$ (both verify at $x = 6$: 270 and 12). **3.** (a) $30x - 10$ (b) $-9x - 21$ (c) $15xy + 20xz$ (d) $-x^2 + 6xy$ (e) $4x - 12y + 8z$ (f) $-6ab + 3ac - 12ad$ (g) $5x^2 + 20x$ (h) $3x^2 - 12xy + 6xz$. **4.** (a) $4x + 19$ (b) $-12x - 15$ (c) $7x - 15$ (d) $18x + 2$ (e) $-12x + 22$ (f) $9x - 12$ (g) 12. **5.** (a) $4x \cdot 3x = 12x^2$, not $12x$; $12x^2 + 4xy$ (b) $x^3 \cdot x^2 = x^5$, not x^6; $x^5 - 3x^4 + 5x^3$ (c) sign error: $-3(x - 2) = -3x + 6$; -6. **6.** (a) $-5a^2 + 17a$ (b) $2x^2 + 11$ (c) $2x + 10$ (d) $w^3 + 3w - 9$ (e) $39x + y$ (f) $-18x^4 - x^3$. **7.** (a) $44a^2 + 14a$ (b) $9x^2 + 11x$. **8.** C. **9.** C. **10.** 3. **11.** (a) $x^2 + 10x + 24$ (b) $6x^2 + 19x + 10$ (c) $5y^2 - 27y - 18$ (d) $a^4 - 5a^2 - 36$. **12.** (a) $2x^2 + 5x + 3$ (b) $x^2 + x - 6$. **13.** (a) $2x + 1$ and $x - 2$ (b) $2x^2 - 3x - 2$. **14.** (a) 748 (b) 2 583 (c) 1 566 (d) 5 776. **15.** (a) $x^2 + 7x + 12$ (b) 182 (c) 13×14 is $(x + 3)(x + 4)$ with $x = 10$. **16.** (a) $x^2 + 15x + 54$ (b) $4a^2 + 29a - 24$ (c) $pq - 9p - 5q + 45$ (d) $x^2 + 4xy - 21y^2$. **17.** (a) $x^2 + 12x + 32$ (b) $y^2 + 4y - 45$ (c) $m^2 - 14m + 33$ (d) $6x^2 - 13x - 28$ (e) $5a^2 + 28ab - 12b^2$ (f) $9y^2 - 42y + 49$. **18.** Area $= (4a + 3)(a + 6) = 4a^2 + 27a + 18$ cm^2 (all three methods agree). **19.** (a) $10x^2 + 29x - 21$ (b) $9h^2 - 9h - 4$ (c) $4z^2 - 25$ (d) $2x^4 + 2x^2y^2 - 12y^4$. **20.** 372 ($p = 3$, $q = 7$, $r = 2$). **21.** 4.

Lesson 3 Assignment

1. (a) $x^2 - 12x + 36$ (b) $x^2 - 121$ (c) $16x^2 - 8xy + y^2$ (d) $9x^2 + 30xy + 25y^2$ (e) $16x^2 - 9$ (f) $9y^2 - 12y + 4$ (g) $25p^2 + 40p + 16$ (h) $36m^2 - 25n^2$ (i) $49a^2 - 28ab + 4b^2$ (j) $64 - 9x^4$ (k) $25a^2 - 90ab + 81b^2$ (l) $9a^6 - 16$. **2.** (a) $24x^2 - 66x + 15$ (b) $60x^2 + 4x - 8$ (c) $-6a^2 - 50a - 84$ (d) $75x^2 + 60x + 12$ (e) $90x^2 - 10y^2$ (f) $-18a^2 - 60ab + 48b^2$. **3.** (a) $2x^2 - 2x + 46$ (b) $6x - 12$ (c) $2x^2 + 32$ (d) $-6xy + 6y^2$. **4.** (a) $-x^2 - 3x + 4$ (b) $30x - 18$ (c) $3x^2 - 22x + 4$ (d) $-50x^2 + 56x + 70$ (e) $38x^2 + 40x - 2$ (f) $7p^2 + 13pq + 19q^2$. **5.** (a) $2x^2 + 16x + 34$ (b) $7a^2 - 32ab - 15b^2$ (c) $-8y^2 - 4y + 13$ (d) $-x^2 - 20x + 49$ (e) $-12y^2 + 34y$. **6.** A (only $(x + 4)^2 = x^2 + 16$ is incorrect; it should be $x^2 + 8x + 16$). **7.** (a) $x^2 + 6x + 9$ m^2 (b) $6x + 9$ m^2 (c) $64 - 25 = 39$ and $6(5) + 9 = 39$. **8.** (a) both sides equal 4 (b) at $x = 2$, $9x^2 + 16 = 52 \neq 4$; Jen left out the middle term $-24x$. **9.** *(Extension)* 6.

Lesson 4 Assignment

1. (a) $p^3 - p^2 - 7p - 20$ (b) $2n^3 - 3n^2 - 14n + 15$. **2.** (a) $x^3 - 8x^2 + 19x - 42$ (b) $3a^3 - 11a^2 - 38a - 24$ (c) both sides equal -28. **3.** (a) FOIL gives only 4 of the $2 \times 3 = 6$ products; Marco missed $x \cdot x = x^2$ and $2 \cdot x = 2x$ (b) $x^3 + 3x^2 + 3x + 2$. **4.** (a) $3x^5 - 13x^3 - 9x^2 - 10x + 45$ (b) $-n^3 - 4n^2 - n + 6$ (c) $3a^3 - 14a^2b + 3ab^2 + 20b^3$ (d) $24x^3 + 30x^2 - 51x - 45$. **5.** (a) $2x^3 + 17x^2 + 38x + 15$ (b) $6h^3 - 37h^2 + 37h - 10$ (c) $9a^3 + 18a^2b - 16ab^2 - 32b^3$ (d) $6x^3 - 19x^2y - 65xy^2 + 50y^3$. **6.** $V = (3z + 2)^3 = 27z^3 + 54z^2 + 36z + 8$. **7.** $V = (3x + 2)^2(x - 4) = 9x^3 - 24x^2 - 44x - 16$. **8.** (a) $-8b^4 + 4b^3 - 20b^2 + 12b + 12$ (b) $-3x^4 + x^3 + 15x^2 + 7x - 4$ (c) $-x^3 + 21x^2 - 39x + 30$. **9.** 18. **10.** B (the square $(b + 3)^2$ is $b^2 + 6b + 9$, not $b^2 + 9$ — the middle term $6b$ was dropped in Line 2).

Lesson 5 Assignment

1. (a) $a = 3$, $b = 12$, $c = -8$; area = 127 (b) $a = 7$, $b = 20$, $c = 10$; area = 55.75. **2.** (a) $a = -2$, $b = 1$, $c = 54$ (b) area = 33. **3.** (a) $x^2 + 60x + 900$ cm^2 (b) $x^2 + 60x$ cm^2 (c) 18.09 cm^2. **4.** (b) $14x^2 - 7$ m^2. **5.** (a) outer radius $= (s + t)$ mm (b) $A = \pi(s + t)^2 - \pi s^2 = 2\pi st + \pi t^2$. **6.** (a) $y = -3$ (b) $t = 1$ (c) $m = 5$ (d) $n = \frac{10}{7}$. **7.** $x = 5$; sides 8 cm, 15 cm, 17 cm. **8.** (a) $15a^2 + 14a - 8$ cm^2 (b) length = 15.5 cm, width = 14.5 cm (c) side = 13 cm. **9.** (a) $2x^3 + 7x^2 - 19x + 6$ cm^2 (b) 33.5 cm^2. **10.** (a) $8x^3 - 12x^2 + 6x - 1$ mm^3 (b) 91.125 cm^3. **11.** (a) $16x^3 - 4x^2 - 36x + 9$ cm^3 (b) $40x^2 - 8x - 18$ cm^2 (c) volume = 297 cm^3, surface area = 318 cm^2. **12.** B. **13.** 2.8. **14.** 139 cm.

Lesson 6 Assignment

1. C. **2.** B. **3.** A. **4.** B. **5.** A. **6.** D. **7.** D. **8.** C. **9.** 24 $(4x + 20)$. **10.** 336. **11.** (a) $6x^2 - 7x - 20$ (b) $3x^3 + 10x^2 - 23x + 10$ (c) at $x = 2$ both sides equal 28. **12.** (a) $2x - 1$ and $x + 3$ (b) $2x^2 + 5x - 3$ (one shaded and one unshaded x-tile cancel). **13.** *(Extension)* 94 $(p = 9, q = 4)$.